

Introduction to Health Economics

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- In the previous analysis, we assumed both the insured individual and the insurance company knew in advance the individual's probability of sickness and could set premiums and payouts accordingly.
- In the real insurance market, insurance firms and customers do not share identical knowledge.
 - Firms, which view customers from a distance, might have trouble judging who is likely to stay healthy and who is likely to get sick.
 - Customers familiar with their own medical history and unhealthy habits have intimate knowledge of their own risk.
- Asymmetric information: a situation in which agents in a potential economic transaction do not have the same information about the quality of the good being transacted.
- Used-car market: Asymmetric information with the seller is common in this market.
 - Used cars and insurance contracts are very different products. However, the lessons about asymmetric information from the used-car market can be applied readily to insurance markets.

- Symmetric information:
 - Imagine a well-functioning used car market.
 - Sellers advertise cars, and buyers can accurately assess the condition of each car for sale
 - Buyers and sellers have symmetric info about car quality - symmetric information
 - Outcome: each car sells for a different price, depending on its quality
- Pareto-improving transaction: a transaction that make at least one person better off without making any other individual worse off.
- One goal of a market is to make sure all Pareto-improving transactions take place.
- In the market we have described, there is nothing to stop all Pareto-improving transactions from taking place
 - All the cars end up with the people who value them the most

- Nobel prize-winning economist George Akerlof imagined what would happen if the used-car market described above suffered from information asymmetry (Akerlof 1970).
 - Sellers can determine car quality, but buyers cannot - asymmetric information
 - All cars look identically good to the buyers
- Asymmetric information in this market will lead to different outcomes in this market:
 - Any car that sell will sell for the same price (say P). Why?
 - Inability to differentiate the quality of car will force the seller offering car at $P'(>P)$ to lower his price to P to have a chance of finding a buyer.
 - Not all cars reach the market. Why?
 - Sellers who own high quality cars worth more than P will not want to put their cars on sale at all.
 - Outcome: only the lower-quality cars stay on the market - first example of adverse selection.
- Adverse selection: the oversupply of low-quality goods, products, or contracts that results when there is asymmetric information. For instance, if a supplier of a product has better information about product quality than a buyer, then the highest-quality products will not be offered.

- Summary:
 - Cars only sell at one price P .
 - As a result, the high quality cars worth more than P leave the market.
- What do buyers do?
 - Buyers know the cars remaining on the market is of low quality. Thus, the average value of the remaining cars is less than P .
 - Buyers thus know that they would be purchasing a car worth much less than P , so unless the market price declines, buyers will refuse to purchase any cars.
- Suppose a new price P'' establishes itself below P ($> P''$).
 - Again, the same process will drive out the cars worth more than $P'' \Rightarrow$ average worth of the cars remaining on the market falls further $\Rightarrow$ buyers respond by refusing to purchase.
- With each new price drop and subsequent round of adverse selection, the car quality in this market continues to degrade until only the lowest quality goods are still on the market.
- Even those may not sell if the utility function of buyers requires some level of quality higher than the existing ones in the market.

- The market unravels, and potential Pareto-improving transactions do not occur. This is a market failure.
- Hence, market for used cars can fail under asymmetric information due to information advantage with the sellers and the incentive to misrepresent the quality of their cars.
 - Sellers of high quality cars cannot guarantee to buyers the quality of the cars they are selling $\Rightarrow$ Sellers of low-quality cars have incentive to mimic as sellers of high quality cars
- Is there any solution to this market failure?
 - What if buyers value cars so much that they are willing to pay top dollar for even the lowest-quality cars?
 - What would be the effect of a law that forbade low-quality cars from being placed on the market?
- Let's introduce a formal statement of the Akerlof model to see whether the market functions under these conditions?

- Both sellers and buyers seek to maximize their own utility.
- Before any transaction, each buyer and seller owns a set of cars, which is labelled 1 through n.
- Each car has exactly one owner.
- Sellers and buyers derive utility from the cars they own and other goods named as M expressed in INR.

$$U_S = \sum_{j=1}^n X_j + M \quad (1)$$

$$U_B = \sum_{j=1}^n \frac{3}{2} X_j + M \quad (2)$$

- where X_j - quality of the jth car owned and M - utility derived from other goods
- Buyers value cars 50 percent more than sellers because buyers naturally desire cars more than sellers do.

- Car quality X is uniformly distributed between 0 and 100.

$$X \sim \text{Uniform}(0, 100) \quad (3)$$

- Cars are equally likely to have any quality level between 0 and 100

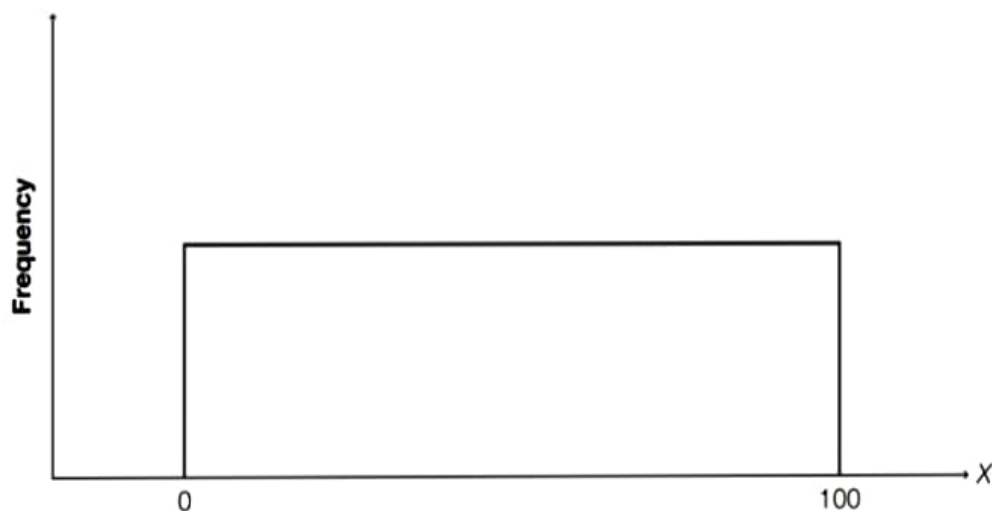


Figure 1: The distribution of car quality.

- Buyers do not know the true quality of a particular car but they know the utility function of the sellers and know the distribution of cars available for sale.
- Buyers also understand that sellers will withdraw highest-quality cars if the price does not justify selling.
- All the cars must share the single price P as the cars in this market are indistinguishable to buyers.
- A seller will put a car on the market if selling it will increase his utility.

$$U_S(\text{before}) = \sum_{j=1}^n X_j + M \quad (4)$$

- Suppose that a seller sells car 1 with quality level X_1 .

$$U_S(\text{after}) = \sum_{j=1}^n X_j - X_1 + M + P \quad (5)$$

- Change in seller's utility is $P - X_1$
- Hence, the seller will only put car 1 on the market if $P \geq X_1$.

- It means the car j is offered on the market only if the quality level X_j is less than or equal to the single price P . Price P serves as an upper bound on the quality of cars offered in the market.
- Let $\Omega(P)$ be the set of all cars still on offer in the market at price P , which coincides with the set of cars with quality less than or equal to P

$$\Omega(P) = (i | X_i \leq P) \quad (6)$$

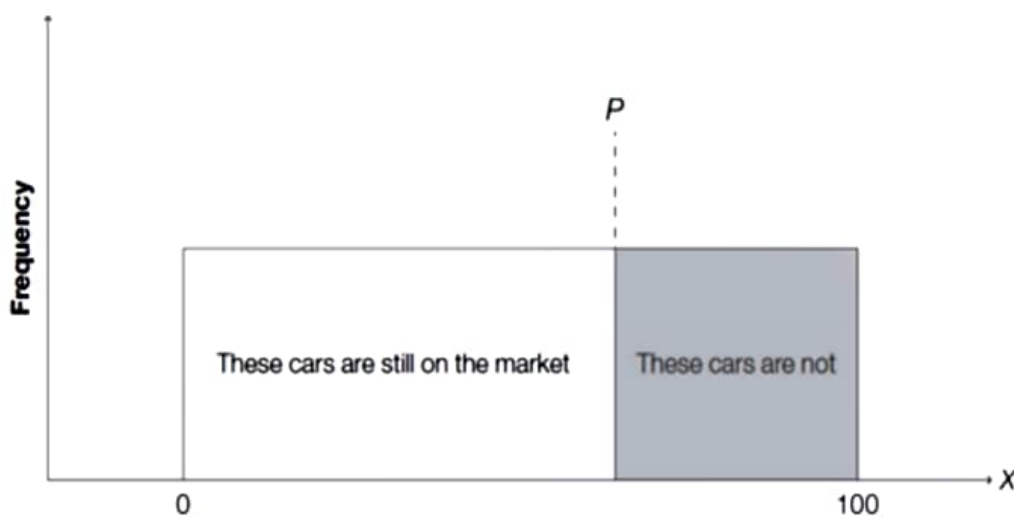


Figure 2: Adverse selection in the used-car market.

- Like sellers, buyers also try to maximize utility. However, the quality of the car a buyer gets is uncertain.
- Buyers know that the car is drawn from a distribution of $\Omega(P)$ (the set of cars remaining on the market after sellers have withdrawn the cars whose quality exceeds P)
- Buyers have to think in terms of expected utility.

$$U_B(\text{before}) = \sum_{j=1}^n \frac{3}{2} X_j + M \quad (7)$$

- Suppose that a buyer buys a car. It will be her $(n + 1)$ th car with quality level X_{n+1} .

$$U_B(\text{after}) = \sum_{j=1}^n \frac{3}{2} X_j + \frac{3}{2} X_{n+1} + M - P \quad (8)$$

- Unfortunately, the buyer does not know the actual value of X_{n+1} , and hence cannot compute $U_B(\text{after})$.
- However, using $\Omega(P)$ she can evaluate her expected utility from buying the car.

- Change in buyer's utility is $\frac{3}{2}E[X_{n+1}] - P$
- Hence, buyers will buy if $\frac{3}{2}E[X_{n+1}] \geq P$.
- The expected marginal benefit of the purchase (the additional expected utility from car $n + 1$ with quality X_{n+1}) must outweigh the marginal cost of buying it, P .
- At price P , quality of the remaining cars is uniformly distributed between 0 and P .
- As car owners with the quality of car better than P have already left. Now the car will be drawn from a distribution of $\Omega(P)$.
- $E[X_{n+1} | P] = \frac{1}{2}P$
- Using this condition into the condition for buying $\frac{3}{2}E[X_{n+1}] \geq P$ gives $\frac{3}{4}P \geq P$
- This is impossible; hence buyers will not buy for any P .
- No cars sell, no Pareto-improving trades take place, the cars stay with sellers (who do not want them as much as the buyers do). The market unravels.

- Summary

- A single price P is somehow established in the market because of asymmetric information.
- Sellers remove all cars of quality greater than P .
- Of the cars that remain, the average quality $E[X|P]$ is only $\frac{1}{2}P$.
- Buyers do not like cars enough to buy a car of expected quality $\frac{1}{2}P$ for a price of P .
- No cars sell, even though buyers like cars better than sellers and all the cars “should” end up with buyers.

- Let's link used car market with health insurance market.
- Each customer i has an expected amount of health care costs over the course of the year X_i .
- An insurance company offers a single policy with an annual premium P . This full insurance policy covers all health care costs incurred during the year.
- Customer i will purchase insurance if and only if P is less than his expected health care costs X_i .
- The insurers cannot distinguish healthy and sick customers.
- Expected customer health care costs X_i are distributed uniformly in the population between \$0 and \$20,000.

- Analogy between these two markets:
 - The “cars” are customers’ bodies.
 - The “sellers” are customers
 - The “buyers” are insurance companies
 - The sellers try to convince the buyers that the “bodies” are healthy; just as a high-quality car is worth a lot to buyers, a healthy customer is worth a lot to insurers.
 - Just like high-quality cars leave the market when a universal price is set, high-quality bodies will leave the market when a universal premium is set.
- Suppose the insurer offers a contract with premium \$10,000 for the year.
- What happens? Who stays in the market?

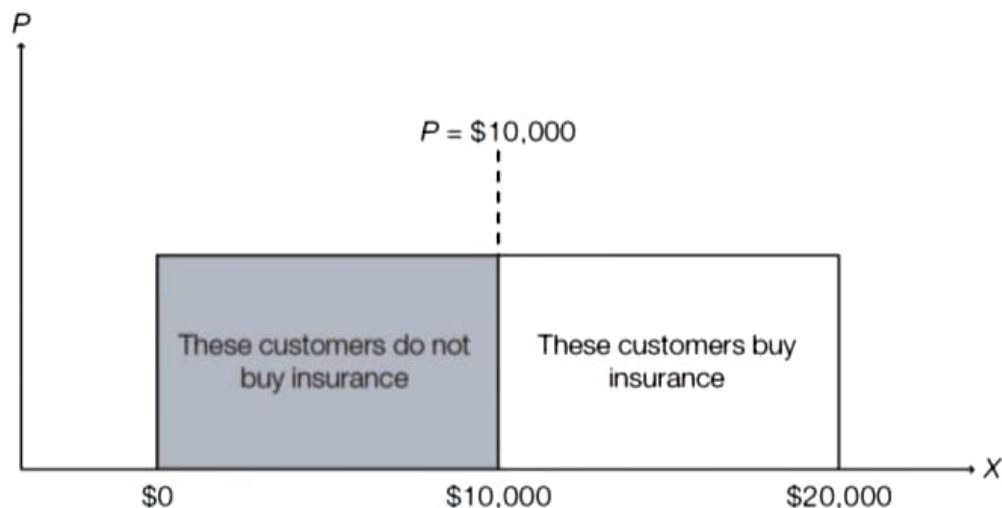


Figure 3: Adverse selection in the health insurance market.

- Only the least healthy people buy insurance; their average health expenditures are \$15,000.
- The insurer raises premiums to \$15,000 the next year.

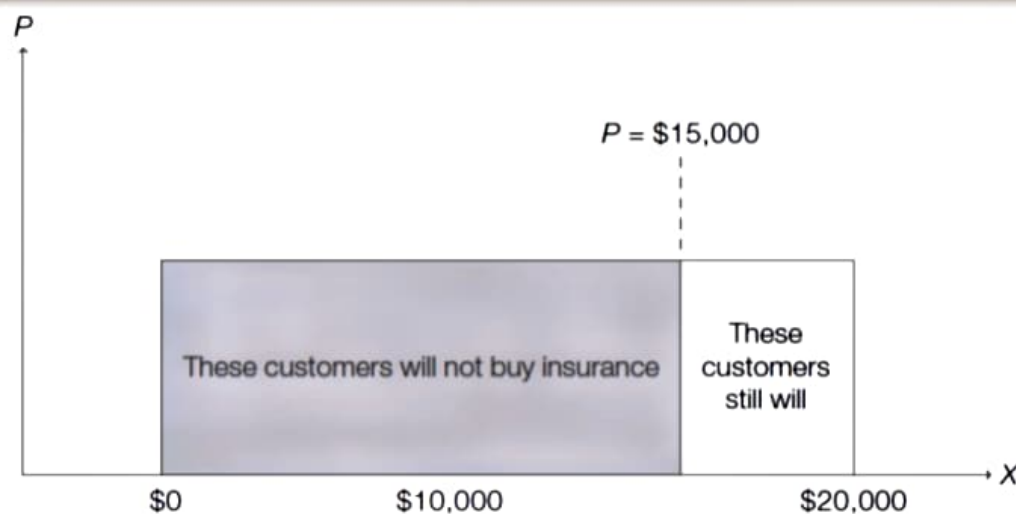


Figure 4: Adverse selection in the health insurance market.

- There is nothing to stop this cycle, which is called an adverse selection death spiral.
- Adverse selection death spiral: successive rounds of adverse selection that destroy an insurance market.
- The heart of the problem is adverse selection: only the worst customers stay in the market when the insurer sets the premium.
- No way for the insurer to turn a profit in this very simple model.

- When can the market for lemons work?
- What if buyers value cars very highly?

$$U_S = \sum_{j=1}^n X_j + M \quad (9)$$

$$U_B = \sum_{j=1}^n \frac{5}{2} X_j + M \quad (10)$$

- Now buyers value cars much more than sellers. Will this fix the market?
- New condition for buyers to buy:
- $\frac{5}{2} E[X_{n+1}] \geq P$
- Recall that $E[X | P] = \frac{1}{2} P$. This is unaffected by the buyers' utility function – why?
- The last two conditions imply buyers will be willing to buy cars at price P . Even though they know the remaining cars are bad. Why?
- They value them highly enough to pay P for them.

- What if there is a minimum guaranteed car quality?

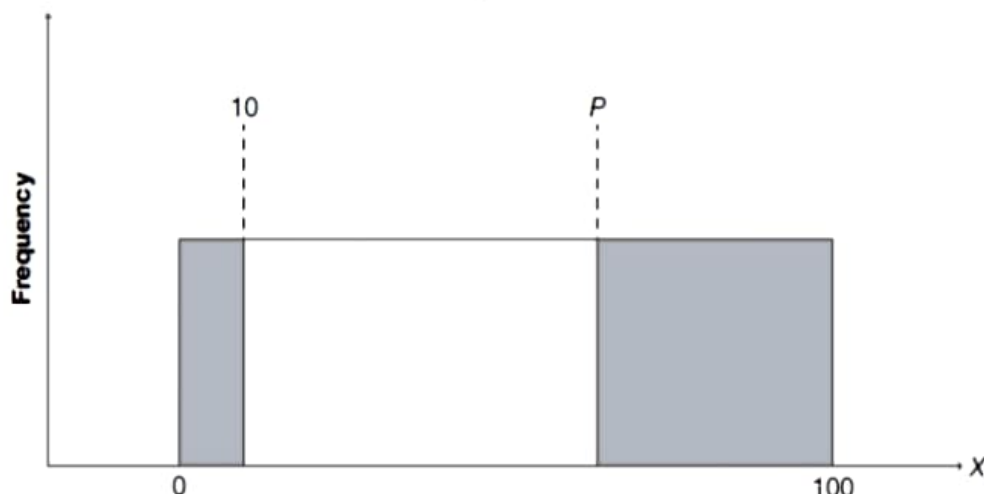


Figure 5: Adverse selection in the used-car market with a guaranteed minimum car quality.

- The worst cars were forced off the market, so the remaining cars are better.
- The condition for buyers ($\frac{3}{2}E[X_{n+1}] \geq P$) is as it was before, but now $E[X | P]$ will be different because a different subset of cars is on the market. Find $E[X | P]$.
- Buyers buy if the price is below 30.

- What if the government sets a price ceiling?
- Suppose a consumer-protection agency decides to set a price ceiling in the market for lemons in an attempt to protect buyers from being gouged by unscrupulous sellers.
- What effect does this price ceiling have on the market?
- The answer is almost none.
- The only thing it affects is $\Omega(P)$ – the set of cars that sellers bring to the market.
- Sellers cannot charge more than the ceiling price, so suppose that sellers charge some price less than the ceiling.
- They will withhold the top-quality cars just as before – in fact, the price ceiling ensures that the best cars (those more valuable than the price ceiling) never reach the market under any circumstances.
- Buyers know this and, as before, value the typical car on the market less than the price, and the market collapses again.

- In some cases, ceiling price may actually harm the market.
- Suppose instead that buyers value any given car 150 percent more than sellers, as in the previous example.

$$U_B = \sum_{j=1}^n \frac{5}{2} X_j + M \quad (11)$$

- In this case, buyers were always willing to purchase cars, almost regardless of the price.
- On the contrary, the price ceiling forces sellers to leave the highest-quality cars off the market.
- Those cars may have found buyers who would have benefited from a sale, even at a high price.
- Hence in this case, a well-meaning reform might lead to economic inefficiency.